A Robot Bird That Can Fly and Swim

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Atlantic puffins do more than fly. They also use their wings to swim underwater and chase food. Now, scientists at MIT have built a robot bird that does the same thing. The team was led by Raphael Zufferey.

The robot is called the FAAV. It uses its wings to move between air and water. That is hard because water is about 1,000 times **denser** than air, so moving through water takes much more force. The team gave the robot bendy wings that **flex** up to 90 degrees in water. They tested three wing sizes; medium wings worked best. The wings were also **coated** with tiny bits that push water away, so they stay light.

In tests, the robot swam at nearly 3 feet (1 meter) per second and flew at about 20 feet (6 meters) per second, as fast as real diving birds. It took off from the water using only its wings, without **paddling** its feet. The team shared its work in the journal Science on July 9, 2026. One day, the robot could help scientists study the ocean.

Word Treasure

Atlantic puffins -- A kind of seabird that can fly in the air and also swim underwater using its wings.

denser -- Packed more tightly together, so something feels heavier for its size.

flex -- To bend without breaking.

coated -- Covered with a thin layer of something.

paddling -- Moving feet back and forth in the water, like rowing.

Background you need

Some real seabirds, like puffins, use their wings to both fly in the air and swim underwater.

Building a machine that can do both is hard because water is much thicker than air, so wings need a special shape to work in both places.

Scientists copy ideas from nature to build robots that may one day help explore the ocean.

Break it down

WHO: Scientists at MIT, led by Raphael Zufferey
WHAT: They built a robot bird called the FAAV that can fly and swim using its wings
WHERE: In tests in the air and in the water
WHY: To copy how puffins move, and one day help scientists study the ocean

Why it matters

It shows how copying animals can help inventors build machines that explore places you may one day study or visit.

Test what you remember

1. What is the robot bird called?

- (A) Puffin
- (B) FAAV
- (C) Science
- (D) MIT

2. Which real bird inspired the robot?

- (A) Eagle
- (B) Atlantic puffin
- (C) Owl
- (D) Penguin

3. Why is moving between air and water hard for the robot?

- (A) The ocean is too cold
- (B) The wings are too small
- (C) The robot is too heavy to fly
- (D) Water is about 1,000 times denser than air

4. Which wing size worked best in the tests?

- (A) Small wings
- (B) Medium wings
- (C) No wings at all
- (D) Large wings

5. How did the robot take off from the water?

- (A) By paddling its feet
- (B) Using only its wings
- (C) By jumping off a rock
- (D) With a rocket boost

6. How fast did the robot fly in the tests?

- (A) About 20 feet (6 meters) per second
- (B) About 3 feet (1 meter) per second
- (C) About 1,000 feet per second
- (D) About 90 feet per second

Answer key (try the quiz first!): 1: B 2: B 3: D 4: B 5: B 6: A

Questions to think about

MY THOUGHTS (at least 20 words):